PATTERN CLUSTERING WITH KOHONEN SELF-ORGANIZING MAPS (SOM) WORD

CLUSTERING USING COMPETITIVE LEARNING

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1. Introduction

This report presents the implementation and evaluation of a Kohonen Self-Organizing Map (SOM) for the task of word clustering using Competitive Learning. Self-Organizing Maps are a type of unsupervised neural networks introduced by Teuvo Kohonen, designed to cluster high-dimensional data into meaningful groups through a self-organizing process.

The objective of this assignment is to cluster words belonging to four semantic categories: Animals, Countries, Fruits, and Vegetables. The dataset consists of pre-trained GloVe word embeddings, where each word is represented by a 300-dimensional feature vector that captures semantic meaning and contextual similarity between words.

The SOM model is based on the concept of Competitive Learning, where neurons compete to become the winning neuron, known as the Best Matching Unit (BMU). During training, the winning neuron and its neighboring neurons are updated in order to organize similar patterns into nearby regions within the map.

In this assignment, different values of the parameter k, representing the number of neurons or clusters, were tested from 2 to 10 to analyze the clustering performance under different SOM configurations. The SOM algorithm was implemented entirely from scratch using Python and NumPy without relying on external SOM libraries.

The clustering performance was evaluated using two external evaluation metrics: Purity and Rand Index. In addition, contingency tables and graphical visualizations were used to analyze and compare the clustering results for different values of k.

2. Dataset

The dataset used in this assignment consists of four text files, where each file represents a specific semantic category: animals, countries, fruits, and vegetables (veggies). Each line in the files contains a word followed by its 300 dimensional GloVe embedding represented as space-separated numerical values. The embeddings are used as feature vectors for the SOM clustering process.

The dataset contains the following categories:

- Animals: 50 words
- Countries: 161 words
- Fruits: 58 words
- Vegetables (Veggies): 60 words

The total number of words in the dataset is 329. Each word is associated with a ground-truth semantic category label used for evaluating the clustering performance using Purity and Rand Index metrics.

During data loading, the program checks that each word embedding contains exactly 300 numerical features before adding it to the dataset. Invalid vectors with incorrect dimensions are ignored to ensure consistency in the input data representation.

3. SOM Implementation

3.1 Algorithm Descriptiontion

A Kohonen Self-Organizing Map (SOM) is an unsupervised neural network used for clustering high-dimensional data using Competitive Learning. In this model, each neuron contains a weight vector with the same dimensionality as the input data.

The training process begins by initializing the SOM weights using randomly selected samples from the dataset with small random noise added to the weights in order to improve the initial distribution. The network is then trained over multiple epochs using all input vectors.

During each training step:

- An input vector is presented to the network.
- The Euclidean distance between the input vector and all SOM neurons is calculated.
- The Best Matching Unit (BMU), which is the neuron with the minimum distance to the input vector, is identified.
- The BMU weights are updated to move closer to the input vector using the learning rate.

The weight update process is based on Competitive Learning, where only the winning neuron is updated during training. In addition, the learning rate gradually decreases across the epochs to improve training stability.

The following settings were used in the implementation:

- Number of epochs: 70
- Initial learning rate: 0.8
- k values tested: from 2 to 10
- Random seed: 50

The SOM algorithm was implemented entirely using Python and NumPy without relying on external SOM libraries, in accordance with the assignment requirements.

3.2 Cluster Assignment

After the training process is completed, each input word is assigned to its corresponding Best Matching Unit (BMU). The BMU represents the neuron with the minimum Euclidean distance to the input vector. As a result, words with similar semantic representations are grouped into the same cluster.

The cluster assignment process is performed by calculating the Euclidean distance between each input vector and all SOM neurons, then assigning the word to the neuron with the smallest distance. The predicted clusters are later used to build contingency tables and evaluate the clustering performance using Purity and Rand Index metrics.

4. Evaluation Metrics

4.1 Purity

Purity is an external evaluation metric used to measure the quality of clustering results by determining how well each cluster contains data points from a single ground-truth category.
In this assignment, Purity is used to evaluate how accurately the SOM model groups semantically similar words into the correct categories. According to the assignment description, higher Purity values indicate better clustering performance.

$$\text{Purity} = \frac{1}{n}\sum_{q=1}^{k}\max_{j} n_q^j$$

where n represents the total number of words, k represents the number of clusters, and n_qj^ represents the number of words belonging to class j inside cluster q.
The Purity value ranges from 0 to 1, where values closer to 1 indicate more accurate clustering results.

4.2 Rand Index

The Rand Index (RI) is another external clustering evaluation metric used to measure the similarity between the predicted clustering results and the ground-truth labels. It evaluates clustering performance by measuring the proportion of correctly grouped and correctly separated pairs of data samples.

$$\text{RI} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}$$

where:

- TP represents pairs assigned to the same cluster and belonging to the same class.
- TN represents pairs assigned to different clusters and belonging to different classes.
- FP represents pairs assigned to the same cluster but belonging to different classes.
- FN represents pairs assigned to different clusters but belonging to the same class.

The Rand Index ranges from 0 to 1, where a value closer to 1 indicates stronger agreement between the clustering results and the actual categories.

5. Results

5.1 Summary of Evaluation Scores

k (Neurons)	Purity	Rand Index
2	0.6717	0.8262
3	0.8237	0.9346
4	0.9453	0.9638
5	0.9422	0.8561
6	0.9392	0.8469
7	0.9422	0.7986
8	0.9514	0.8088
9	0.9574	0.8398
10	0.9422	0.7969

The table below summarizes the Purity and Rand Index values obtained for each value of k used during SOM training. Different values of k, ranging from 2 to 10, were tested in order to analyze the clustering performance under different SOM configurations.

These results are used to compare the quality of the generated clusters and identify the most suitable value of k based on the evaluation metrics.

5.2 Evaluation Plot

Figure 1 shows the Purity and Rand Index scores obtained from the SOM model for different values of k ranging from 2 to 10. The graph illustrates how the clustering performance changes as the number of SOM neurons increases.

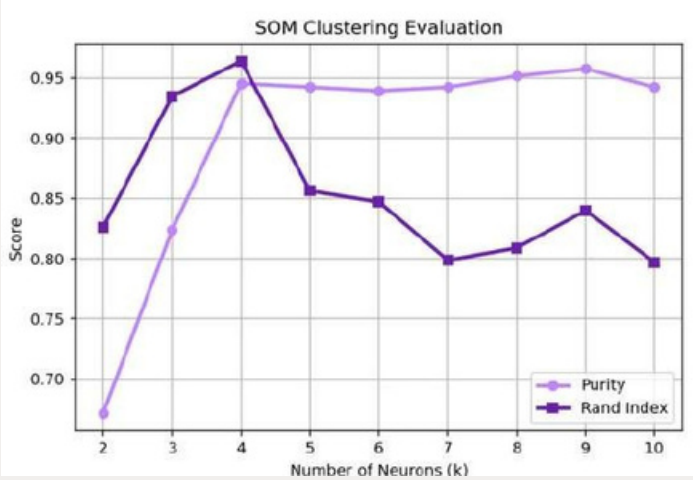


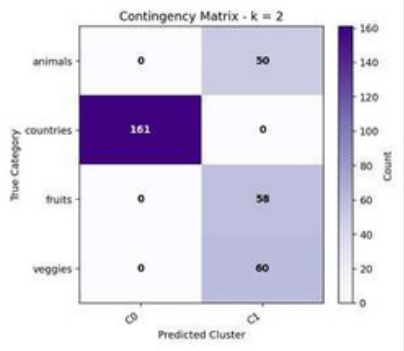
Figure 1: Purity and Rand Index scores for different SOM neuron values (k = 2 to k = 10).

6. Contingency Tables

The contingency matrices show the distribution of the ground-truth semantic categories across the clusters generated by the SOM model for different values of k. The rows represent the true word categories, while the columns represent the predicted SOM clusters. These matrices are used to analyze the clustering quality and evaluate how effectively the SOM model groups semantically similar words into the same cluster.

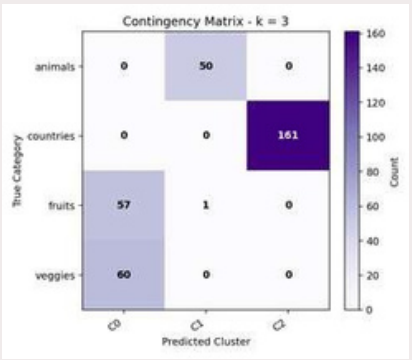
k = 2 Neurons | Purity = 0.6717| Rand Index = 0.8262

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	0	161	0	0
Cluster 1	50	0	58	60



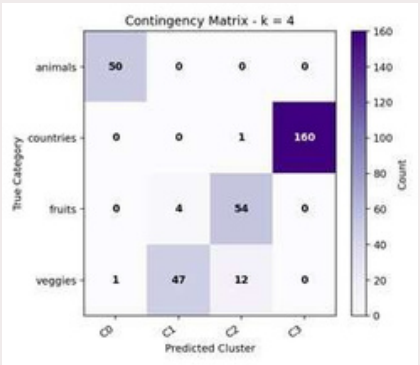
k = 3 Neurons | Purity = 0.8237| Rand Index = 0.9346

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	0	0	57	60
Cluster 1	50	0	1	0
Cluster 2	0	161	0	0



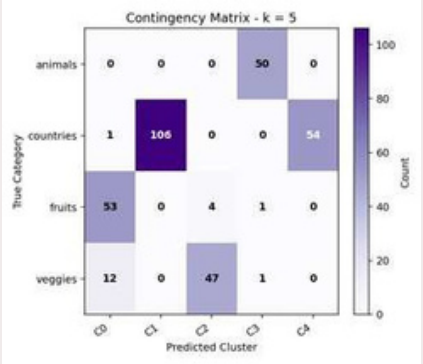
k = 4 Neurons | Purity = 0.9453| Rand Index = 0.9638

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	50	0	0	1
Cluster 1	0	0	4	47
Cluster 2	0	1	54	12
Cluster 3	0	160	0	0



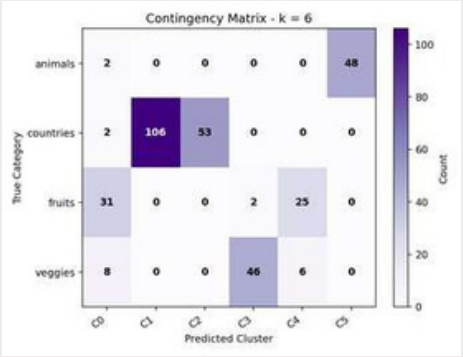
k = 5 Neurons | Purity = 0.9422| Rand Index = 0.8561

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	0	1	53	12
Cluster 1	0	106	0	0
Cluster 2	0	0	4	47
Cluster 3	50	0	1	1
Cluster 4	0	54	0	0



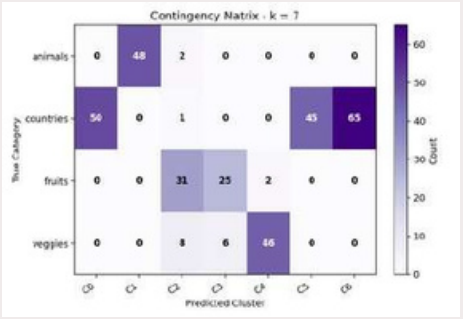
k = 6 Neurons | Purity = 0.9392| Rand Index = 0.8469

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	2	2	31	8
Cluster 1	0	106	0	0
Cluster 2	0	53	0	0
Cluster 3	0	0	2	46
Cluster 4	0	0	25	6
Cluster 5	48	0	0	0



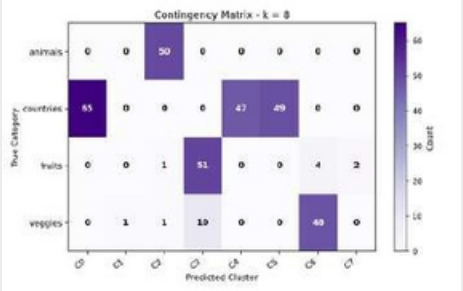
k = 7 Neurons | Purity = 0.9422| Rand Index = 0.7986

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	0	50	0	0
Cluster 1	48	0	0	0
Cluster 2	2	1	31	8
Cluster 3	0	0	25	6
Cluster 4	0	0	2	46
Cluster 5	0	45	0	0
Cluster 6	0	65	0	0



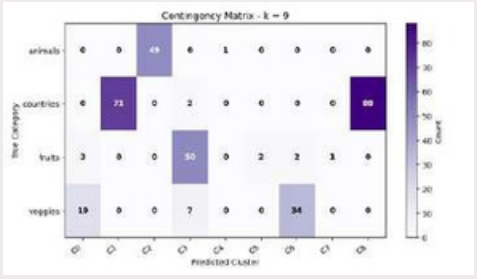
k = 8 Neurons | Purity = 0.9514| Rand Index = 0.8088

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	0	65	0	0
Cluster 1	0	0	0	1
Cluster 2	50	0	1	1
Cluster 3	0	0	51	10
Cluster 4	0	47	0	0
Cluster 5	0	49	0	0
Cluster 6	0	0	4	48
Cluster 7	0	0	2	0



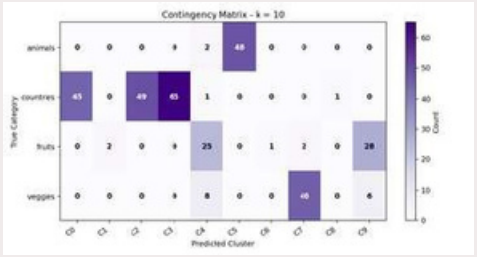
k = 9 Neurons | Purity = 0.9574| Rand Index = 0.8398

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	0	0	3	19
Cluster 1	0	71	0	0
Cluster 2	49	0	0	0
Cluster 3	0	2	50	7
Cluster 4	1	0	0	0
Cluster 5	0	0	2	0
Cluster 6	0	0	2	34
Cluster 7	0	0	1	0
Cluster 8	0	88	0	0



k = 10 Neurons | Purity = 0.9422| Rand Index = 0.7969

Cluster	Animals	Countries	Fruits	Veggies
Cluster 0	0	45	0	0
Cluster 1	0	0	2	0
Cluster 2	0	49	0	0
Cluster 3	0	65	0	0
Cluster 4	2	1	25	8
Cluster 5	48	0	0	0
Cluster 6	0	0	1	0
Cluster 7	0	0	2	46
Cluster 8	0	1	0	0
Cluster 9	0	0	28	6



7. Discussion

7.1 Effect of k on Purity

The Purity score generally improved as the number of SOM neurons increased from k = 2 to k = 9. The highest Purity value was achieved at k = 9 with a score of 0.9574, indicating that the SOM was able to form highly homogeneous clusters with minimal category mixing.

At lower values of k, especially k = 2 and k = 3, multiple semantic categories were grouped into the same clusters, which reduced clustering purity. As k increased, the SOM allocated more neurons to represent the semantic structure of the dataset more effectively.

For k = 4, which matches the actual number of categories in the dataset, the model achieved strong performance with Purity = 0.9453 and Rand Index = 0.9638. The contingency matrix shows that most categories were clearly separated, particularly the Countries and Animals categories.

When k became larger than the true number of categories, some categories were divided into multiple smaller clusters. This increased Purity slightly, but the Rand Index gradually decreased because the clustering became more fragmented.

7.2 Effect of k on Rand Index

The Rand Index reached its highest value at k = 4 with a score of 0.9638. This result is reasonable because the dataset contains four true semantic categories, allowing the SOM to produce clusters that closely match the ground-truth labels.

At lower values of k, several categories were grouped together in the same clusters, which reduced the agreement between the predicted clusters and the true labels. As k increased beyond 4, the SOM began dividing some categories into smaller sub-clusters. Although this slightly improved Purity, it caused the Rand Index to decrease because words from the same category were distributed across multiple clusters.

These results demonstrate the trade-off between Purity and Rand Index. Larger k values improve cluster homogeneity, while k values closer to the true number of classes provide better agreement with the actual category labels.

7.3 Cluster Interpretability

The contingency matrices show that the SOM model was generally successful in separating the semantic categories across different values of k. In most experiments, the Countries category formed highly distinct clusters compared to the other categories, indicating that country word embeddings possess more distinct semantic characteristics.

In contrast, the Animals, Fruits, and Veggies categories showed a higher degree of overlap in some clusters. This behavior is expected because these categories share similar semantic and contextual relationships within the GloVe embedding space.

At higher values of k, the SOM model was able to create more specialized clusters with less category overlap, which contributed to the improvement in Purity scores observed in the evaluation results.

7.4 Optimal k

Based on the evaluation results, the highest Purity value was achieved at k = 9 with a score of 0.9574, while the highest Rand Index was achieved at k = 4 with a score of 0.9638.

The value k = 4 represents the actual number of semantic categories in the dataset, which allowed the SOM model to produce clusters that closely matched the ground-truth labels. On the other hand, larger values of k produced more specialized and homogeneous clusters, but increased the fragmentation of some categories.

Therefore, the choice of the optimal k depends on the evaluation objective. If the goal is to maximize agreement with the true categories, k = 4 is the best choice. However, if the goal is to obtain highly homogeneous clusters, k = 9 provides the highest Purity.

8. Conclusion

In this assignment, a Kohonen Self-Organizing Map (SOM) was implemented from scratch for the task of word clustering using 300-dimensional GloVe embeddings.

The SOM model was trained using different values of k ranging from 2 to 10 and evaluated using Purity and Rand Index metrics.

The experimental results demonstrate the ability of the SOM network to organize semantically similar words into meaningful clusters using Competitive Learning. In addition, contingency matrices and evaluation metrics were used to analyze the clustering performance under different SOM configurations. Overall, this assignment demonstrates the effectiveness of SOM as an unsupervised neural network for clustering high-dimensional word embeddings and analyzing semantic relationships between words.

9. References

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